

CHARUSAT

DEVANG PATEL INSTITUTE OF ADVANCE TECHNOLOGY AND RESEARCH
PROJECT SYNOPSIS

Date: / /

MAJOR PROJECT(CE451.01/CSE403/IT461) 2025-26

Project Title:	Decentralized CVE Management Platform Using Hyperledger Fabric - Secure Vulnerability Disclosure & Governance
Alternate Title	1. Permissioned Blockchain CVE Registry & Coordination 2. Distributed Vulnerability Management Network 3. Fabric-Based CVE Numbering Authority (CNA) Network
Project Goal	To develop a permissioned Hyperledger Fabric blockchain platform that decentralizes CVE (Common Vulnerability & Exposure) coordination, replacing centralized MITRE model with immutable vulnerability ledger, role-based access (CNAs vs public), embargoed disclosure, lifecycle management, and consortium governance.
Objectives	1. Design multi-org Fabric network for CVE Numbering Authorities (CNAs), national bodies, and regulator participation with role-based channels. 2. Implement Go chaincode enforcing CVE schema (ID, product, severity, CVSS score, affected versions, embargo timestamp, status transitions). 3. 3. Develop REST APIs for CNA submission, public query, NVD indexing, and event emission for off-chain systems. 4. Create React dashboard for CVE lifecycle tracking (draft → published → deprecated), severity assessment, and embargo management. 5. Implement DAO-style governance voting for CVE project prioritization and disclosure policy evolution.
Project Brief description	Hyperledger Fabric permissioned network replacing centralized CVE coordination with decentralized governance. Multi-org setup: CVE Numbering Authorities (CNAs), national agencies, security researchers, regulatory bodies. Go chaincode enforces CVE record schema, lifecycle state machine (draft/published/deprecated), role-based access (CNA-only write, public read), embargoed disclosure windows, atomic assignment. REST API + React dashboard for CVE submission, querying, timeline visualization. Event-driven architecture for NVD synchronization. Governance voting on disclosure policies.

Expected outcome	Company/Govt: Trustless CVE coordination, reduced assignment delays, enhanced transparency. Customer/Researcher: Public immutable CVE ledger, clear disclosure timelines, consortium-driven policy evolution. Developer: Advanced Fabric patterns (RBAC, private data, governance), cybersecurity domain expertise, production enterprise blockchain.			
Project Category:	Blockchain + Cybersecurity + Governance			
Rate Project yourself	5 out of 5. Why: Novel academic research application, addresses real MITRE/NVD limitations, high-impact (global CVE ecosystem), responsible disclosure, governance + security combination, published research potential.			
Development Environment	Software: Ubuntu 22.04, Docker 20.10+, Docker-Compose 2.0+, Node.js 16+, npm 8+, Hyperledger Fabric 2.4+, Go 1.19+, React 18+, PostgreSQL 14+. Hardware: 8GB+ RAM, 60GB+ SSD, multicore CPU. Team: Same (YES). Integration: Monorepo (fabric/ + api/ + dashboard/ + research/ folders), automated docker-compose, GitHub Actions CI/CD.			
Execution/Hosting Environment	Software: Ubuntu 22.04, Docker 20.10+, Kubernetes 1.24+, PostgreSQL 14+, Redis (caching), Prometheus (metrics). Hardware: Cloud: AWS EKS 3x t3.large (16GB RAM, 100GB storage); On-prem: 3x bare-metal (32GB RAM, 200GB SSD). Hosting: YES, cloud K8s deployment with REST API + React dashboard. Public-read blockchain access via API gateway.			
Statistics	Time Duration: 6 Months Team Size: 1 Developer Division of Responsibility: Developer 1: Fabric network design, multi-org MSP setup, Go chaincode (CVE schema, lifecycle, RBAC), governance contracts. Node.js REST API, database design (PostgreSQL), event listeners, NVD sync logic, security audit. React dashboard (CVE submission form, timeline viz, governance voting), frontend auth (JWT), performance optimization.			
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